M2 Progress Report – Faithful RAG: Evaluating Citation Accuracy and Retrieval Robustness in Domain-Specific Scientific Literature

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Team Faithful RAG

1 Progress Summary

Three of the four proposal components run end-to-end on a 37-question gold benchmark of intellectual-property and innovation-policy papers. The retrieval ablation grew from four to sixteen configurations (new corpus-level SOTA); the faithfulness verifier exposes a 55–61 pp gap between NLI and LLM-judge labels; the RAGAS pipeline pits chunked RAG against a 1M-token long-context baseline; the Corrective RAG hook is wired and awaits threshold ablations in M3.

2 Gold Benchmark

Each team member contributed ∼10 question-passage pairs across policy impact, methodology, comparison, factual extraction, and multi-hop reasoning. A pair encodes an atomic claim, the source `paper_id`, character-span offsets on a line-feed-normalised `document.md`, and a verbatim quote. A validator byte-checks every span against its quote, and CI rejects fabricated concatenations. Inter-annotator agreement uses five pairs per annotator labelled *supports*/*contradicts*/*unrelated*; six adversarial controls (q900–q905) populate the rejection cells and break the single-class Cohen-κ paradox (Cohen, 1960) that the first IAA round triggered.

3 Retrieval Ablation

The ablation crosses four embedder families (OpenAI `text-embedding-3-small`; BGE-M3 (Chen et al., 2024); E5-large (Wang et al., 2024); ColBERTv2 (Santhanam et al., 2022)), two chunk granularities (coarse ∼2000 chars, fine ∼300 chars), and an optional ZeroEntropy cross-encoder reranker. Sixteen configurations share one `HybridRetriever` code path and report hit@k, MRR, and nDCG over n=37 paper-level gold pairs.

E5-large with the reranker wins every paper-level metric and beats OpenAI even without it (+10.8 pp hit@10, +8.6 pp MRR): a 335M-parameter open-source encoder edges out the closed-source baseline. ColBERTv2 trails E5-large by 17 pp MRR despite a more expressive per-token similarity; its MS-MARCO pretraining fails to transfer to scholarly text. ColBERTv2 is the only family that prefers fine chunks. Multi-hop questions saturate hit@10 at 1.000 on coarse and fall to 0.667–0.833 on fine across every family, locating the chunker as the dominant bottleneck.

config (`coarse_rerank`)	hit@5	hit@10	MRR
OpenAI `text-emb-3-small`	0.946	0.946	0.842
BGE-M3	0.946	0.946	0.842
E5-large	**0.973**	**0.973**	**0.878**
ColBERTv2	0.892	0.919	0.786

Table 1: Best configuration per embedder family. Full 16-config grid lives in the repo `retrieval_eval/results`.

4 Faithfulness Verification

The verifier extends the proposal's NLI pipeline (DeBERTa-v3-large-mnli, He et al., 2021) with an LLM-judge head. Three generators (GPT-4o-mini, Claude-3.5-haiku, DeepSeek-chat) answer the 24 answerable gold questions closed-book; both verifiers label each (claim, gold-passage) pair.

generator	NLI	LLM-judge	gap
gpt-4o-mini	0.750	0.138	+0.612
claude-3.5-haiku	0.750	0.173	+0.577
deepseek-chat	0.750	0.198	+0.552

Table 2: Fraction labelled *faithful* by each verifier (n=24).

NLI labels three-quarters of hallucinated closed-book answers as faithful at every provider; the LLM judge flags them. The 55–61 pp gap is stable across generators and replicates a known NLI failure mode (Min et al., 2023; Gao et al., 2023). NLI alone fails as a faithfulness gate on this benchmark.

5 End-to-End RAG vs Long-Context

Two pipelines run through the four RAGAS metrics (Es et al., 2024): chunked RAG (the §3 retriever, GPT-4o-mini) and a long-context baseline that loads every cited `document.md` into a 1M-token Gemini 2.5 Flash window.

metric	RAG	long-context	Δ
faithfulness	0.806	0.975	+0.169
answer_relevancy	0.662	0.799	+0.138
context_precision	0.979	1.000	+0.021
context_recall	0.792	1.000	+0.208

Table 3: RAGAS metrics, n=8 stratified subset.

Long-context wins every metric. The widest gaps fall on context_recall (+20.8 pp) and faithfulness (+16.9 pp): chunked RAG hides ∼17 pp of available factual support behind chunk boundaries. Two metrics point at one root cause and reinforce the chunker bottleneck from Table 1.

6 Limitations and M3 Plan

Three scope ceilings remain: RAGAS at n=8, partial chunk-level coverage (16–20 of 37 queries), and a closed-book versus retrieved-context split between the two faithfulness verifiers. M3 delivers four fixes. (i) A no-gap sliding-window chunker: the multi-hop fine-chunk failures, partial chunk coverage, and the +17 pp long-context advantage share one root cause. (ii) Domain-adaptation of ColBERTv2 on synthetic contrastive pairs from the gold set, sized to close the 17 pp MRR gap. (iii) The full 37-question RAGAS sweep across all 16 retriever configurations. (iv) Activation of the wired CRAG (Yan et al., 2024) scorer as a 17th configuration aimed at the multi-hop `fine` failures isolated in §3.

References

Jianlv Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. 2024. BGE M3-Embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation.

Jacob Cohen. 1960. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1):37–46.

Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. 2024. RAGAS: Automated evaluation of retrieval augmented generation.

Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. 2023. Enabling large language models to generate text with citations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*.

Pengcheng He, Jianfeng Gao, and Weizhu Chen. 2021. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing.

Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*.

Keshav Santhanam, Omar Khattab, Jon Saad-Falcon, Christopher Potts, and Matei Zaharia. 2022. ColBERTv2: Effective and efficient retrieval via lightweight late interaction. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics*.

Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, and Furu Wei. 2024. Text embeddings by weakly-supervised contrastive pre-training.

Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, and Zhen-Hua Ling. 2024. Corrective retrieval augmented generation.